

Datadive GDGoC '26

Instructions & Guidelines

Event Overview

Duration	6:00 PM – 10:00 PM
Submission Format	.ipynb file (code and markdown), 1-page pdf summary and optionally deployed app link
Communication	Announcements in #server-announcements, and queries in #main
Team Size	Solo
Judging	Post-datathon, participants will be updated on their email

Dataset

You will work with the APS (Air Pressure System) Failure at Scania Trucks dataset.

The dataset has anonymized features (eg aa_000, aa_001 etc). These features represent various readings. Most features are counters, although binned variables from histograms exist in this dataset as well.

The “class” feature has the following values which mean:

- Pos: Truck failure is due to APS issues
- Neg: Truck failure is due to a different issue, unrelated to APS

NOTE: Neg does not mean the truck is “Healthy”

Problem Statement

A Scania service center receives trucks that have already experienced some component failure. Diagnosing whether the failure originates from Air Pressure System (APS) in comparison to some other system failure saves diagnostic time of physical inspection.

You are a Data Scientist assigned to solve this problem. You must predict the truck’s root cause of failure, specifically if it’s APS failure or otherwise. This is a binary classification problem.

Scania also publishes an asymmetric cost matrix for this problem:

Error Type	Cost
False Positive (FP)	Cost A = 10
False Negative (FN)	Cost B = 500

Total Cost = (Cost A * number of FP) + (Cost B * number of FN)

You must work on your model with this cost function optimization as well. You are required to select an appropriate decision threshold and justify it.

You must test your model against the test dataset.

Deliverables

1) Jupyter notebook (.ipynb) which must include:

- Exploratory Data Analysis: missingness patterns, class distribution and appropriate visualizations of the data
- Feature Engineering: Selecting appropriate features and manipulating them into a more actionable form
- Model: A classification ML model of your choice and appropriate metrics for its testing
- Threshold Justification: Choose your threshold for the classification and justify it
- Business Translation: All your extracted insights and predictions must be written in markdown cells in a viable format

2) 1-page PDF summary involving key insight, final metrics, threshold rationale

3) (Optional) Link to your deployed app (streamlit etc)

AI Policy

You are allowed to use AI for general coding, commenting and formatting

However, your insights must be original and unique. Your reasoning should be clear and concise. The quality of your analysis holds the highest weightage.

Evaluation Rubrics

Criteria	Description	Weightage
EDA	Depth and Correctness of Exploratory Data Analysis	30%
Insights	Quality/originality of written insights	30%
Modelling	Predictive model performance & appropriateness	25%
Code Quality	Clarity and quality of the notebook's code	15%

** Brownie points for deployment*